

Master 4 Grid Maker

Official User Manual

Purpose

Turn one whole-degree geographic square into a repeatable 10 x 10 system of 100 exact map extents. The tool also creates a numbered EPSG:3857 GeoTIFF overlay and a production manifest for QGIS batch mapping.

Requirements

Windows, Python 3, and Pillow.

```
py -3 --version
```

```
py -3 -m pip install Pillow
```

Keep these together

Master4_Grid_Maker.py

Master4_Grid_Maker.bat

Master4_Grid_Maker_User_Manual.pdf

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Run it

Double-click Master4_Grid_Maker.bat. Enter four whole-degree edges in this exact order: west, east, south, north.

82w, 81w, 30n, 31n

The box must be exactly 1 degree wide by 1 degree high and inside the practical EPSG:3857 latitude range.

Outputs

Files are written to C:downloads:

Master4_82W_81W_30N_31N_Extents.txt

Master4_82W_81W_30N_31N_Grid.tif

100-cell address

Use the first decimal digit of absolute latitude as the tens row. Use the first decimal digit of absolute longitude plus 1 as the column. Do not round.

30.56N, 81.34W -> cell 54

30.59999N, 81.30001W -> cell 54

30.500001N, 81.399999W -> cell 54

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Production manifest

Each row contains the cell number, exact QGIS EPSG:3857 extent, and deterministic TPKX filename.

```
01 -9028010.7033, -9016878.7543, 3503549.8435, 3516410.3983 [EPSG:3857]
```

```
W082W081N031N030-001-GHY-Z20.tpkx
```

The filename identifies the Master 4 and cell, so cells can be selected in any order without sequential batch naming.

Reference overlay

Load the generated _Grid.tif in QGIS or ArcGIS Earth. It displays the 01-100 cells directly over the map; background value 0 is tagged NoData.

Worldwide addressing

The numbering logic is hemisphere-aware and works in W/N, E/N, W/S, and E/S boxes within practical Web Mercator coverage.

QGIS batch workflow

1. Generate the Master 4 for the degree square you want.
2. Load the grid TIFF over your basemap.
3. Choose the numbered cells you need; skip cells you do not need.
4. Copy each exact extent into QGIS batch processing.
5. Use the manifest filename as that map's permanent geographic identity.
6. Produce raster MBTiles with your locked production settings.
7. Convert finished MBTiles to TPKX with mb2tpkx when ArcGIS Earth is the target viewer.

Troubleshooting

Invalid coordinate: use a number followed by N, S, E, or W.

Wrong span: edges must be exactly one whole degree apart.

Non-whole-degree edge: all four Master 4 edges must be whole degrees.

Pillow is required: install it with the command on page 1.

Quick reference

Input: west, east, south, north

Output: 100 manifest rows + numbered GeoTIFF

Address: first decimal latitude digit + longitude digit plus 1

No rounding.